

Hawkeye EQMS — Process-Flow Analysis

Per-module lifecycle, persona × action × API × screen mapping, AI integration. Source-of-truth document for the test plan and demo scripts.

Document version	v2.0
Revised on	2026-04-24
Revised by	Hawkeye Engineering
Revision note	Adds AI-assist mapping, persona × action matrix, and detailed state transitions verified against live code.
Modules covered	5 (Deviation / Non-Conformance · CAPA — Corrective & Preventive Action (v2) · Document Control (SOP / WI / Policy / Spec) · Risk Register (FMEA) · Management Review (MRM))
Total lifecycle states	43
Total legal transitions	32
AI assists wired in	10

Contents

- Deviation / Non-Conformance
- CAPA — Corrective & Preventive Action (v2)
- Document Control (SOP / WI / Policy / Spec)
- Risk Register (FMEA)
- Management Review (MRM)
- Cross-cutting infrastructure

1. Deviation / Non-Conformance

Compliance	ISO 9001:2015 §10.2 · 21 CFR 211.192 · ICH Q10 §3.2.3
Purpose	Capture, investigate, disposition and close any planned or unplanned deviation from a specification, SOP, batch record, or quality system requirement. Drive CAPA when systemic.
Model file	backend/src/models/DeviationModel.js
Collection	deviations
Numbering	DEV-YYYY-NNNN
Classification	CRITICAL MAJOR MINOR

LIFECYCLE STATES

State	Primary owner	Description
REPORTED	QA Specialist (admin)	Initial intake; raised by detector via 'Report Deviation' dialog. Captures title, description, classification, type, category, area, processStep, productName, batchNumbers, immediateActions.
UNDER_ASSESSMENT	QA Specialist / Head of QA	Impact assessment under way: product quality, patient safety, batch disposition, regulatory impact.
UNDER_INVESTIGATION	QA Specialist / SME	Root-cause analysis. Method may be 5-Why / Fishbone / FMEA / FTA. AI 5-Why scaffold available.
PENDING_DISPOSITION	Head of QA	RCA complete; awaiting batch disposition decision (RELEASE / REJECT / REWORK / REPROCESS / QUARANTINE).
PENDING_CAPA_DECISION	Head of QA	Disposition done; deciding whether systemic CAPA is required.
CAPA_REQUIRED	QA / CAPA Owner	CAPA required and being created (auto via cross-module service or linked manually).
PENDING_CLOSURE	Head of QA / VP Quality	All actions closed; final review before formal closure.
CLOSED	VP Quality (tenant_admin)	Terminal. closedBy + closedAt + closureNotes captured. Triggers e-sig if configured.
CANCELLED	Head of QA	Terminal. Raised in error; cancel reason captured.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
REPORTED	UNDER_ASSESSMENT	Assess	POST /api/deviations/:id/assess	productQualityImpact, patientSafetyImpact, batchDisposition, regulatoryImpact	EDITOR_ROLES (admin/buyer/auditor/tenant_admin/insp

From	To	UI button	API	Required input	Allowed roles
REPORTED UNDER_ASSESSMENT	UNDER_INVESTIGATION	Investigate	POST /api/deviations/:id/investigate	method, summary, rootCause, rootCauseCategory	EDITOR_ROLES
UNDER_INVESTIGATION	PENDING_DISPOSITION	Investigate (sets rootCause)	POST /api/deviations/:id/investigate (with rootCause)	rootCause set	EDITOR_ROLES
PENDING_DISPOSITION	PENDING_CAPA_DECISION	Dispose	POST /api/deviations/:id/dispose	decision, justification	EDITOR_ROLES
PENDING_CAPA_DECISION	CAPA_REQUIRED PENDING_CLOSURE	(decision drawer)	POST /api/deviations/:id/capa-decision	capaRequired (bool); optional autoCreateCapa	EDITOR_ROLES
CAPA_REQUIRED PENDING_CLOSURE	CLOSED	Close	POST /api/deviations/:id/close	closureNotes	EDITOR_ROLES
(any)	CANCELLED	Cancel	DELETE /api/deviations/:id (soft)	n/a	EDITOR_ROLES

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
5-Why Scaffolder (Wave 1)	POST /api/ai/deviation/scaffold-five-why	UNDER_INVESTIGATION	View+AI drawer · 'Scaffold 5-why with AI' popover	Generates a 5-level why chain with citations + 6M (Man/Machine/Method/Material/Measurement/Environment) categorisation. Free Gemini.
CAPA RCA Drafter (Wave 1)	POST /api/ai/capa/draft-rca	UNDER_INVESTIGATION → CAPA_REQUIRED	View+AI drawer · 'Draft CAPA RCA' drawer	Drafts severity + RCA + corrective/preventive actions + effectiveness plan. User edits + e-signs before saving.
Predictive CAPA badge (Wave 3)	POST /api/ai/predict/capa-outcome	PENDING_CAPA_DECISION	View+AI drawer (auto-renders)	Heuristic prediction P(on-time) + P(effective) + top contributing factors.
Signal cluster detector (Wave 3)	GET /api/ai/signals?status=open	all states (post-create)	Head-of-QA dashboard	Z-score trend detection on equipment / material / process clusters.

SCREENS

- /nonconformance — List + report + lifecycle actions
- /nonconformance (View+AI drawer) — Detail + AI 5-Why + Draft CAPA RCA + Predictive badge

CLOSURE CONDITIONS

CLOSED requires (a) status set via /close, (b) closureNotes captured, (c) closedBy + closedAt set. If CAPA was required, the linked CAPAs must also reach CLOSED_EFFECTIVE before the deviation is permanently closed.

2. CAPA — Corrective & Preventive Action (v2)

Compliance	ISO 9001:2015 §10.2 · 21 CFR 211.192 · ICH Q10 §3.2.2
Purpose	Formalise root-cause investigation + corrective + preventive actions for a quality issue. Track effectiveness to closure. Enforce 5-stage approval gates (Triage / RCA / Action Plan / Effectiveness / Closure).
Model file	backend/src/models/capaV2Models.js
Collection	capa-v2 + 16 child collections
Numbering	CAPA-YYYY-NNNN

LIFECYCLE STATES

State	Primary owner	Description
DRAFT_CANDIDATE	QA / Auditor	Candidate created from an audit finding or manual draft.
INTAKE_DRAFT	Submitter	Intake form being filled (issue statement, scope, regulatory exposure).
UNDER_TRIAGE	Triage Reviewer (auditor/admin)	Triage decision pending: FORMAL_CAPA / CORRECTION_ONLY / NO_CAPA.
TRIAGE_NO_CAPA	Triage Reviewer	Terminal — no CAPA needed; rationale logged.
CORRECTION_ONLY	CAPA Owner	Quick fix only; no formal investigation.
CAPA_OPEN	CAPA Owner	Formal CAPA opened; ownerUserId + dueDate required.
INVESTIGATION_IN_PROGRESS	Investigator	Investigation scope + activities being recorded.
RCA_PENDING_APPROVAL	Approver (auditor/admin)	Root-cause analysis submitted; awaiting approval.

State	Primary owner	Description
ACTION_PLAN_PENDING_APPROVAL	Approver	Corrective/preventive plan submitted; awaiting approval.
ACTION_PLAN_APPROVED	CAPA Owner	Plan approved; ready to execute action items.
IN_IMPLEMENTATION	Action-item Owners	Action items being worked. Evidence attached as proof.
AWAITING_EFFECTIVENESS_CHECK	Effectiveness Reviewer	Implementation done; awaiting effectiveness evaluation.
EFFECTIVENESS_REVIEW_IN_PROGRESS	Effectiveness Reviewer	Evaluating whether actions achieved their goal (PASS/FAIL).
CLOSED_EFFECTIVE	VP Quality	Terminal. Effective; closure approved.
CLOSED_INEFFECTIVE	VP Quality	Terminal. Ineffective; reopen or supersede needed.
REOPENED	QA / Reviewer	Reopened after recurrence or failed effectiveness.
CANCELLED	Approver	Terminal.
SUPERSEDED	VP Quality	Terminal — replaced by another CAPA.
MERGED	VP Quality	Terminal — merged into another CAPA.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	AI
DRAFT_CANDIDATE	INTAKE_DRAFT UNDER_TRIAGE MERGED CANCELLED	(workspace tabs)	POST /api/capa-v2/intakes (or merge endpoint)	intake fields	m
INTAKE_DRAFT	UNDER_TRIAGE CANCELLED	Submit Intake	POST /api/capa-v2/intakes/:id/submit	all intake fields	m
UNDER_TRIAGE	TRIAGE_NO_CAPA CORRECTION_ONLY CAPA_OPEN	Triage decision	POST /api/capa-v2/triage/:id/decision	decision (NO_CAPA/CORRECTION/FORMAL_CAPA), rationale	tri (ai)
CAPA_OPEN	INVESTIGATION_IN_PROGRESS	Open CAPA	POST /api/capa-v2/capas (then PUT investigation)	ownerUserId, dueDate	m
INVESTIGATION_IN_PROGRESS	RCA_PENDING_APPROVAL	Submit RCA	PUT /api/capa-v2/capas/:id/root-cause	rootCauseStatement, rcaMethod	m
RCA_PENDING_APPROVAL	ACTION_PLAN_PENDING_APPROVAL	Approve RCA	POST /api/capa-v2/capas/:id/approvals (stage=RCA, decision=APPROVED)	approver decision	ap (ai)
ACTION_PLAN_PENDING_APPROVAL	ACTION_PLAN_APPROVED	Approve Plan	POST /api/capa-v2/capas/:id/approvals (stage=ACTION_PLAN, decision=APPROVED)	approver decision	ap (ai)
ACTION_PLAN_APPROVED	IN_IMPLEMENTATION	Start Action	PATCH /api/capa-v2/action-items/:id/status (start any item)	≥ 1 action item moved to IN_PROGRESS	m
IN_IMPLEMENTATION	AWAITING_EFFECTIVENESS_CHECK	Mark Complete	POST /api/capa-v2/capas/:id/implementation-evidence (last action complete)	all action items COMPLETED + evidence	m
AWAITING_EFFECTIVENESS_CHECK	EFFECTIVENESS_REVIEW_IN_PROGRESS	Begin EC	PUT /api/capa-v2/capas/:id/effectiveness (start)	review window dates	m
EFFECTIVENESS_REVIEW_IN_PROGRESS	CLOSED_EFFECTIVE CLOSED_INEFFECTIVE	Close CAPA	POST /api/capa-v2/capas/:id/close	closureOutcome (EFFECTIVE/INEFFECTIVE), closedAt	ap (ai)
CLOSED_EFFECTIVE CLOSED_INEFFECTIVE	REOPENED SUPERSEDED MERGED	Reopen	POST /api/capa-v2/capas/:id/reopen (or supersede)	reason	ap (ai)

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
CAPA RCA Drafter	POST /api/ai/capa/draft-rca	INVESTIGATION_IN_PROGRESS	From deviation View+AI drawer or CAPA workspace	Drafts severity + RCA + actions; user edits + e-signs.
Predictive CAPA Outcome (Wave 3)	POST /api/ai/predict/capa-outcome	CAPA_OPEN onwards	Predictive badge on CAPA row	P(on-time) + P(effective) heuristic.

Agent	Endpoint	Attached to state	Where in UI	What it does
Active Learning Loop	POST /api/ai/active-learning/adjustments	CLOSED_EFFECTIVE/CLOSED_INEFFECTIVE	Drift dashboard	Refines retrieval weights based on accept/reject outcomes.

SCREENS

- /buyer/capas — CAPA workspace (admin/buyer view)
- /auditor/capas — CAPA workspace (auditor view, scoped to assigned)

CLOSURE CONDITIONS

CLOSED_EFFECTIVE requires: all action items COMPLETED, ≥1 implementation evidence, effectiveness check PASS, closure approval (stage=CLOSURE) by approvalRoles user. CLOSED_INEFFECTIVE permitted but triggers reopen workflow.

3. Document Control (SOP / WI / Policy / Spec)

Compliance	ISO 9001:2015 §7.5 · 21 CFR 211.180 · 21 CFR Part 11 §11.10 (e-signature)
Purpose	Author, review, approve, publish, supersede and withdraw controlled documents. Multi-step serial approval. Triggers training auto-assignment when published if requiresTrainingOnUpdate=true.
Model file	backend/src/models/DocumentControlModel.js
Collection	document-controls
Numbering	Sequential per tenant; user-set documentNumber

LIFECYCLE STATES

State	Primary owner	Description
DRAFT	Doc Author	Initial draft; freely editable.
UNDER_REVIEW	Reviewers (per step)	approvalSteps[] active; serial review per step.
APPROVED	Doc Control	All steps APPROVED/DELEGATED; awaiting publish.
EFFECTIVE	(All users)	Live; effectiveDate set. Supersede or withdraw permitted.
SUPERSEDED	Doc Control	Terminal; supersededById points to new revision.
WITHDRAWN	Doc Control	Terminal; withdrawalReason captured.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
DRAFT	UNDER_REVIEW	Submit for Review	POST /api/document-control/:id/submit-for-review	reviewers[] (each with role; userId optional)	authenticated tenant user
UNDER_REVIEW	DRAFT (rejected) UNDER_REVIEW (next step) APPROVED (all approved)	Approve / Reject	POST /api/document-control/:id/approve	stepOrder, decision (APPROVED/REJECTED/DELEGATED), comments	authenticated; step.approverId enforced
APPROVED	EFFECTIVE	Publish	POST /api/document-control/:id/publish	effectiveDate (defaults to today); triggers training auto-assign	doc admin
EFFECTIVE	SUPERSEDED	Supersede	POST /api/document-control/:id/supersede	new doc payload (auto-creates new revision)	doc admin
EFFECTIVE	WITHDRAWN	Withdraw	POST /api/document-control/:id/withdraw	withdrawalReason	doc admin

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
Training Auto-Assign Agent (Wave 2)	POST /api/ai/training/auto-assign-on-sop-revision	EFFECTIVE (publish)	Auto-fired on publish if requiresTrainingOnUpdate=true	Identifies affected roles + creates 'read-and-understood' training records with grace-period dueDate. Optionally drafts an LLM knowledge-check question.

SCREENS

- /document-control — Register + state-action buttons + 'Effective' indicator

CLOSURE CONDITIONS

EFFECTIVE is the operational state. Permanent closure occurs at SUPERSEDED (replaced by new revision) or WITHDRAWN (taken out of service). versionMajor auto-increments on supersede.

4. Risk Register (FMEA)

Compliance	ISO 9001:2015 §6.1 · ICH Q9 (R1) Quality Risk Management
Purpose	Capture process-step risks (failure mode → effect → cause) with FMEA scoring ($S \times O \times D = RPN$). Add mitigations until residual risk acceptable. Tie to source (audit, deviation, complaint, CAPA, regulatory).
Model file	<code>backend/src/models/RiskItemModel.js</code>
Collection	<code>risk-items</code>
Numbering	<code>user-defined</code> / N/A

FMEA RPN BANDS

LOW	RPN < 60
MEDIUM	RPN 60 – 124
HIGH	RPN 125 – 199
CRITICAL	RPN ≥ 200

LIFECYCLE STATES

State	Primary owner	Description
OPEN	Risk Owner	Initial; no mitigations completed.
MITIGATED	Risk Owner	≥ 1 mitigation COMPLETED; residual S/O/D entered; residual RPN computed.
ACCEPTED	VP Quality	Residual risk acceptable; sign-off captured.
CLOSED	VP Quality	Terminal; full audit trail verified.
TRANSFERRED	VP Quality	Terminal; ownership moved to another tenant/site.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
OPEN	OPEN (with mitigation appended)	Add Mitigation	POST <code>/api/risk-items/:id/mitigate</code>	action, owner, dueDate	tenant authenticated
OPEN	MITIGATED	Mark Mitigated	PUT <code>/api/risk-items/:id</code> (status=MITIGATED + residual S/O/D)	≥ 1 mitigation COMPLETED; residual scores	risk owner
MITIGATED	ACCEPTED	Accept Residual	PUT <code>/api/risk-items/:id</code> (status=ACCEPTED)	approver sign-off	tenant_admin
ACCEPTED	CLOSED	Close	PUT <code>/api/risk-items/:id</code> (status=CLOSED)	closure note	tenant_admin
(any non-terminal)	TRANSFERRED	Transfer	PUT <code>/api/risk-items/:id</code> (status=TRANSFERRED)	transfer destination	tenant_admin

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
Risk Scenario Brainstormer (Wave 2)	POST <code>/api/ai/risk/brainstorm-scenarios</code>	OPEN (creation)	Risk Register · 'Brainstorm with AI' button	Given a process description + product class, AI proposes failure modes + effects + causes seeded into draft risk items.

SCREENS

- `/risk-register` — FMEA register with RPN/Band columns + Log Risk dialog

CLOSURE CONDITIONS

CLOSED requires (a) all mitigations COMPLETED, (b) residual RPN acceptable per tenant policy, (c) tenant_admin approval. Residual RPN auto-computed on save when residualSeverity + residualOccurrence + residualDetectability all set.

5. Management Review (MRM)

Compliance	ISO 9001:2015 §9.3 · 21 CFR Part 11 (e-signed minutes)
Purpose	Periodic top-management review of QMS performance: audit results, CAPA status, deviations, complaints, supplier performance, resource adequacy, improvement opportunities, decisions and action items.
Model file	<code>backend/src/models/ManagementReviewModel.js</code>
Collection	<code>management-reviews</code>
Numbering	<code>MR-YYYY-NNNN</code>

LIFECYCLE STATES

State	Primary owner	Description
PLANNED	Chair (VP Quality)	Scheduled; plannedDate + reviewType + chair + attendees set.

State	Primary owner	Description
IN_PROGRESS	Chair / Minutes Taker	Meeting under way; inputs[] being captured (manually or via AI populator).
COMPLETED	Chair + Approver	Terminal; outputs (qmsAdequacy / resourceDecisions / improvementOpportunities / actionItems) captured + e-signed.
CANCELLED	Chair	Terminal.

LEGAL TRANSITIONS

From	To	UI button	API	Required input	Allowed roles
PLANNED	IN_PROGRESS	Start Meeting	PUT /api/management-reviews/:id (status=IN_PROGRESS, actualDate)	actualDate	tenant authenticated
IN_PROGRESS	COMPLETED	Complete + Sign	POST /api/management-reviews/:id/complete	qmsAdequacy, resourceDecisions, improvementOpportunities, actionItems[]	chair / tenant_admin
PLANNED IN_PROGRESS	CANCELLED	Cancel	DELETE /api/management-reviews/:id (soft) or PUT status=CANCELLED	reason	chair

AI ASSISTS WIRED TO THIS MODULE

Agent	Endpoint	Attached to state	Where in UI	What it does
MRM Input Populator (Wave 2)	POST /api/ai/mrm/populate-inputs	IN_PROGRESS	MRM detail · 'Auto-populate inputs' button	Aggregates last N days of audits + CAPAs + deviations + complaints → populates inputs[] with topic/summary/trend. Adds AI narrative.

SCREENS

- /management-review — Register + Schedule Review dialog + detail with inputs + action items

CLOSURE CONDITIONS

COMPLETED requires (a) inputs[] populated for all 9.3.2 topics, (b) qmsAdequacy decision (ADEQUATE/NEEDS_IMPROVEMENT/INADEQUATE), (c) resourceDecisions + improvementOpportunities recorded, (d) actionItems[] with owners + due dates, (e) approvedBy + approvedAt set (e-signature), (f) optionally minutesDocumentId linked.

Cross-cutting infrastructure

Electronic Signatures (21 CFR Part 11)

File: backend/src/routes/electronicSignatureRoutes.js

Endpoint: POST /api/electronic-signatures/sign

Generic e-signature endpoint accepts {recordType, recordId, signatureMeaning, authMethod}. Captures content SHA-256 hash, signer IP, user-agent. Signatures cannot be repudiated; logged to DataIntegrityLog. Used by document-control approvals, MRM completion, CAPA closure.

AI Audit Trail (FDA AI guidance · Jan 2025)

File: backend/src/services/ai/audit/aiAuditTrail.js

Endpoint: Internal: recordAiDecision() · POST /api/ai/decisions/outcome (user accept/reject)

Every AI decision logs feature, prompt-version, retrieval-set hashes, output, confidence, grounded flag, provider+model, latency, accept/reject outcome. Enables drift dashboard + regulator-ready evidence.

Universal Module Config

File: backend/src/middleware/universalPlatform.js

Endpoint: GET /api/universal-platform/config

Each tenant has modules.* boolean flags (EVENT_MANAGEMENT / DOCUMENT_CONTROL / RISK_MANAGEMENT / MANAGEMENT_REVIEW / CAPA / TRAINING / SUPPLIER_QUALITY / etc.). Frontend hides nav and 403s for disabled modules. Novex has all 15 enabled.

Role middleware

File: backend/src/middlewares/roleMiddleware.js

Endpoint: Used as middleware on every protected route

permit(...allowedRoles) gate. Roles normalised: tenantAdmin → tenant_admin, supplierUser → supplieruser, etc. Returns 403 with 'Forbidden: You don't have permission to access this resource.' if role not in allow-list.